

Activity 2

NAME: _____

Q.1. Fully factor the polynomial $9x^2 + 6x + 1 - y^2$.

Q.2. Rationalize the denominator $\frac{16x^2 - y^2}{2\sqrt{x} - \sqrt{y}}$.

Q.3. Simplify the following expression:

$$\frac{(3x+2)^{\frac{1}{2}}(2x+3)^{-\frac{2}{3}} - (2x+3)^{\frac{1}{3}}(3x+2)^{-\frac{1}{2}}}{3x+2}$$

Q.4. Express as a sum of terms of the form ax^r where a is a real number and r is a rational number:

$$\frac{(\sqrt{x}-3)^2}{x^3}$$